

the right triangle with one side of length x and the second side of length y . We can use the Pythagorean theorem to compute the length and use the lengths to compare two vectors' sizes.

[Listing 9-8](#) is a `Vector` class that illustrates the appropriate magic methods for doing math and comparisons between two `Vector` objects. (Each of these methods has additional code using a call to `isinstance()` to ensure that the second object is a `Vector`. These checks are included in the downloadable file, but I have omitted them here to save space.)

File: MagicMethods/Vectors/Vector.py

```
# Vector class

import math

class Vector():
    '''The Vector class represents two values as a vector,
    allows for many math calculations'''
    def __init__(self, x, y):
        self.x = x
        self.y = y

    ❶ def __add__(self, oOther): # called for + operator
        return Vector(self.x + oOther.x, self.y + oOther.y)

    def __sub__(self, oOther): # called for - operator
        return Vector(self.x - oOther.x, self.y - oOther.y)

    ❷ def __mul__(self, oOther): # called for * operator
        # Special code to allow for multiplying by a vector or a scalar
        if isinstance(oOther, Vector): # multiply two vectors
            return Vector((self.x * oOther.x), (self.y * oOther.y))
        elif isinstance(oOther, (int, float)): # multiply by a scalar
            return Vector((self.x * oOther), (self.y * oOther))
        else:
            raise TypeError('Second value must be a vector or scalar')

    def __abs__(self):
        return math.sqrt((self.x ** 2) + (self.y ** 2))

    def __eq__(self, oOther): # called for == operator
        return (self.x == oOther.x) and (self.y == oOther.y)
```